Motion-Based Detection and Tracking of Trypanosoma cruzi Parasites in Microscopy Videos

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Abstract

Chagas disease, caused by the parasite Trypanosoma cruzi, poses a substantial threat to global health, particularly in Latin America. Traditional diagnostics involve the time-consuming and error-prone process of manual analysis of stained blood samples under a microscope. In recent years, the diagnosis process has been significantly enhanced by learning-based algorithms, which analyze digital images of stained blood smears automatically. Nevertheless, computational algorithms can turn ineffective as a result of blood staining being defective by a variety of factors, such as smear preparation, staining, and interpretation. In this collaborative work, we propose an approach for monitoring motile Trypanosoma cruzi parasites in our microscopy videos. The pipeline focuses on movement, contours, and background contrast as primary cues, mainly relying on dense optical flow and a custom point-based tracking algorithm that is both scalable and dependency-free. Code and sample outputs are available at [1].

Keywords: Chagas disease, Fast moving object, Point tracking, Dense optical flow

1 Introduction

Chagas disease, which is caused by the Trypanosoma cruzi (*T. cruzi*) parasite, is a global health problem with a prevalence of millions in Latin America, particularly Mexico. It is transmitted by domestic pets, wild mammals, and insects that feed on blood. Transmission can occur through infected blood transfusions, organ transplants, pregnancy, and contaminated food or drink. Reliable risk assessment, early diagnosis, and optimal treatment strategies are among the challenges that persist, despite the progress that has been made [Rassi and Marin-Neto, 2010, Nagajyothi et al., 2012, Rojo-Medina et al., 2018].

The first learning-based methods for detecting *T. cruzi* in images involve the use of conventional image processing techniques, such as color segmentation, texture analysis, and morphological operations; these methods derive features from microscopic blood smear image samples. Subsequently, these features are processed using machine learning algorithms, including AdaBoost, Support Vector Machines, and Artificial Neural Networks [Soberanis-Mukul et al., 2013, Uc-Cetina et al., 2015]. Deep learning, especially Convolutional Neural Networks (CNNs), has transformed parasite identification in images. CNNs provide end-to-end classification and segmentation solutions without manual feature extraction. Deep learning models can accurately discriminate infected and uninfected blood smear images. These models work directly with raw images to learn hierarchical features [Ojeda-Pat et al., 2022, Pereira et al., 2022, Martins et al., 2023, Rada et al., 2024].

Although identifying the Chagas parasite in videos has shown promising results by using optical flow [Martins et al., 2021, Martins et al., 2024], tracking its movement presents a greater challenge due to its rapid and erratic movement. Here, we enumerate some of the difficulties:

[1] https://github.com/senemaktas/Parasite-Chagas

1. High velocities lead to significant blurring and, large displacement between consecutive frames complicate the position prediction [Rozumnyi et al., 2021, Aktas et al., 2025].

2. The tiny size of the parasite offers limited distinguishability due to low resolution and poor contrast against their background [Haalck et al., 2024].

3. These parasites shows irregularities and transformations in appearance can complicate detection [Chen et al., 2020]

4. Curvilinear trajectories and irregular motions of the parasite can exacerbate the tracking challenges.

Popular tracking libraries, such as DeepSort [Wojke et al., 2017] and StrongSort [Du et al., 2023], perform poorly because *T. cruzi* parasite is tiny and Fast-Moving Object (FMO). We present a method (see Section 3) specifically designed for the automated, label-free detection and tracking of *T. cruzi* parasites in standard microscopy videos (Section 2). The method is designed as a low-cost, easily deployable pipeline that can be integrated into existing microscopy workflows without requiring specialized hardware.

2 Dataset

The dataset collection study was conducted using experimentally infected albino ICR (Institute for Cancer Research) strain mice, which were all in the acute phase of Chagas disease. ICR mice are widely used in scientific research due to their genetic variability, docility, and good reproductive characteristics. Blood samples were collected directly from the tail of each infected animal during the peak of parasitemia. Microscopy was performed using a Motic BA300 microscope, with data acquisition performed at a magnification of 40x. A Canon EOS Rebel T1i, Model 500D digital camera was used for video recording, which was manually controlled by the user rather than through microscopy image analysis software. The camera was connected to the microscope using a C-mount adapter that did not include any optics, and it operated at a resolution of 1280×720 pixels. Three mice were used per experimental replicate. Video recording was performed twice, with a ~30-day interval between sessions. 24 short videos were recorded at 20 FPS, totaling 32 minutes of footage. Some sections of the recordings contain depictions of *T. cruzi*, and some others do not. Furthermore, videos exhibit a wide range of approaches, varying illuminations, and different levels of motion when capturing the parasites. Furthermore, experts extracted 70×70-pixel ROIs from the 24 videos, creating an image dataset of 1,161 positive (parasite present) and 1,697 negative samples.

3 Methodology

An overview of the applied methodology is shown in Figure 1. The experimental methodology steps are explained in the following subsections and have been evaluated on the dataset with the custom metrics.

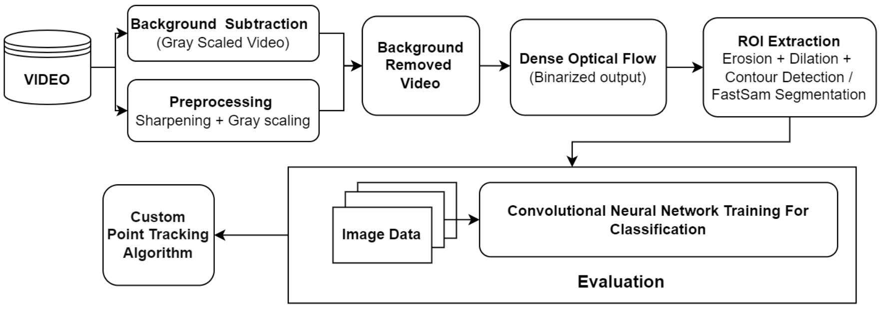


Figure 1: Flowchart of the proposed methodology for monitoring the Trypanosoma cruzi parasite.

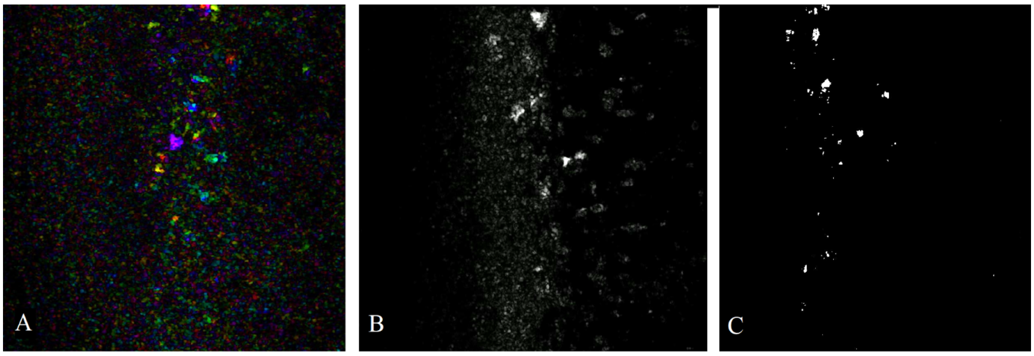


Figure 2: Exploration of dense optical flow output: (A) color shows the direction and magnitude of movement (B) grayscaling preserves motion integrity (C) binarization emphasize potential regions of parasites.

3.1 Pre-processing and Background Subtraction

Preliminary analysis revealed that chromatic information is non-discriminative for parasite localization. So frames are converted to grayscale and convolved with the Laplacian kernel to enhance edges. To avoid per-video parameter tuning under varying backgrounds, we apply background subtraction [Paul et al., 2017]: Algorithm 1 builds an adaptive background, and its absolute difference from each frame isolates moving objects.

Algorithm 1 Background Estimation via Difference-Weighted Averaging

$sum \leftarrow \mathbf{0}$, $count \leftarrow 0$ ▷ running weighted sum of frames, number of accepted frames
for all frames *frame* of video whose index is divisible by frame stride **do**
  if $count = 0$ **then**
    $sum \leftarrow frame$ ▷ seed with the first frame
  else
    $background \leftarrow sum/count$ ▷ current mean background
    $diff \leftarrow (frame, background)/255$
    $weight \leftarrow \mathbf{1} - diff$ ▷ low weight where the frame disagrees
    $sum \leftarrow sum + frame * weight$
  end if
  $count \leftarrow count + 1$
  if $count > maxFrames$ **then break**
end for
return $(sum/count)$

3.2 Dense Optical Flow

The living *T. cruzi* parasites shows rapid movement, distinguishing them from other entities and organisms. Dense optical flow algorithm [Farnebäck, 2003] employed to identify areas of interest where the parasite may reside. Output is visualized as a color-coded image, where each pixel is assigned a specific color that reflects its magnitude and direction.

3.3 Region of Interest (ROI) extraction

To exclude irrelevant objects, morphological operations are applied to the optical flow outputs. Small regions are removed via erosion with 8×8 kernel. Subsequently, broad regions suspected of parasites are enhanced and refined through dilation with 12×12 kernel. Kernel sizes were chosen to avoid over-dilating minor features while preserving candidate parasite regions. Then filtered output is binarized. Figure 2 illustrates the steps.

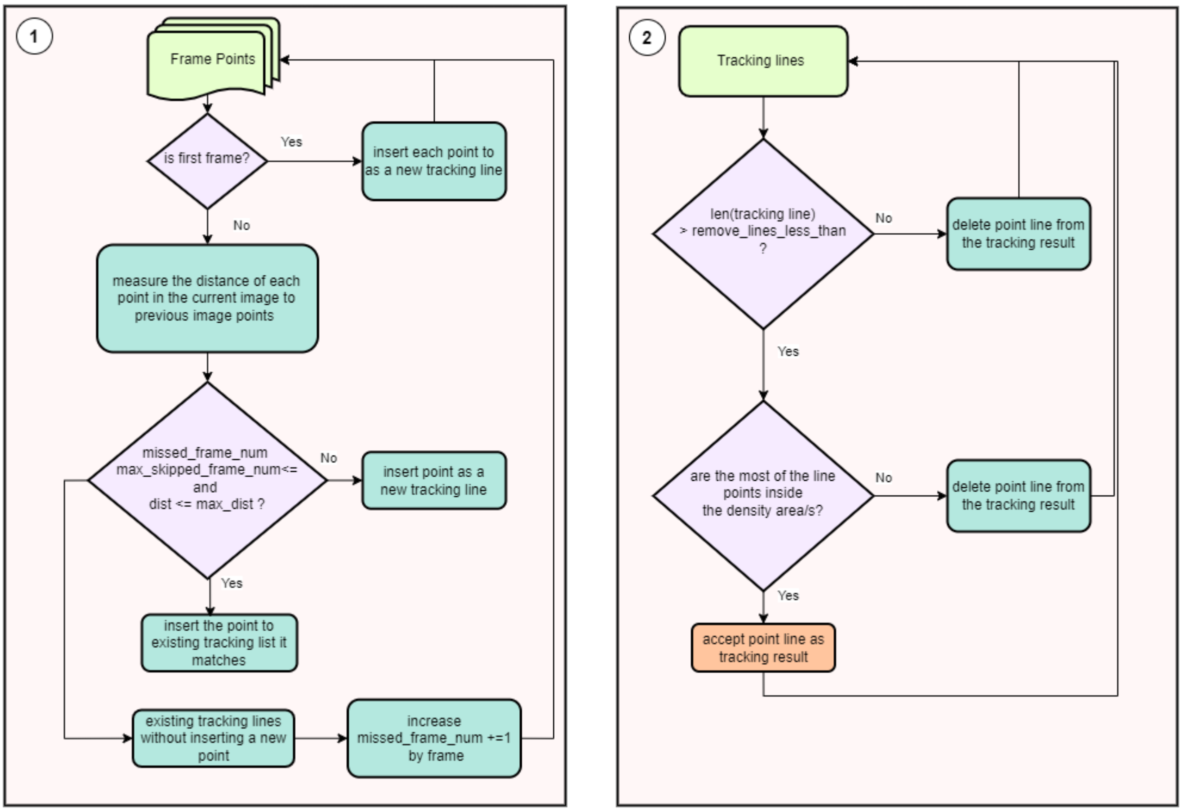
We applied two different methods for localizing parasites. Contour detection [Suzuki et al., 1985] approach is a border-following technique intended to extract contours from binary images. FastSAM [Zhao et al., 2023] algorithm is a simplified variant of the Segment Anything Model that facilitates rapid segmentation of objects in images. Among the two methods, our customized contour detection method tends to identify a wider range of regions, particularly those that may not be optically evident while the FastSAM method seems to adopt a more conservative approach.

3.4 Evaluation

Using image dataset described in Section 2; various CNNs including VGG16 [Simonyan and Zisserman, 2014], InceptionV3 [Szegedy et al., 2016], and Xception [Chollet, 2017] were fine-tuned to confirm selected ROIs whether they truly corresponds to a *T. cruzi* parasite. Our experiments revealed that, aside from VGG16, the other models struggled to effectively classify the ROIs. The VGG16 model was trained for 15 epochs using 70×70 inputs, an SGD optimizer ($\eta = 0.001$), and binary cross-entropy loss with sigmoid activation.

3.5 Custom Point Tracking

To tackle the *T. cruzi* tracking challenge, the proposed technique is depicted in Figure 3.



(a) (b)

Figure 3: Custom Tracking Algorithm Flow. (a) covers the Steps 1 and 2. (b) covers the Steps 3 and 4.

- **Step 1:** In the initial frame, each extracted ROI centre is assigned a unique tracking ID. For each ID, alongside object locations, "*maximum shortest distance*", "*current frame number*", "*start frame number*", and "*end frame number*" values are recorded to facilitate tracking analysis.
- **Step 2:** In the subsequent frames, new objects' points are compared with the recorded tracking points to either align with existing IDs or assign new IDs if no correspondence is found. This step continues until all frames in the video have been thoroughly evaluated.

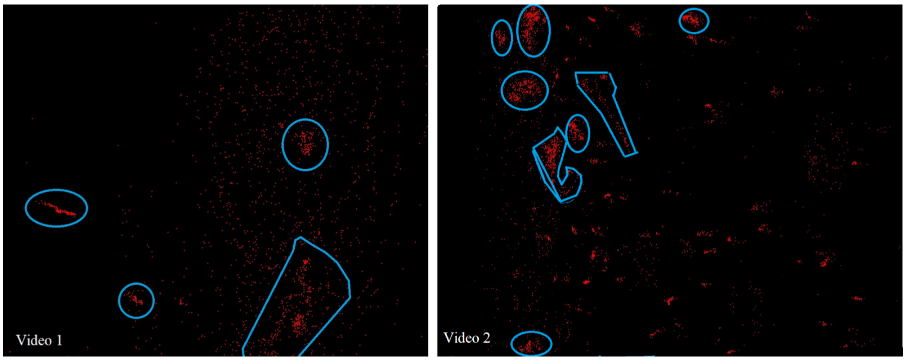


Figure 4: Parasite-dense areas (blue) are mapped by plotting the all ROIs midpoints (red) onto a single image.

1. If the distance between two objects points being compared is less than the "*maximum shortest distance*" variable, the new object point is linked to the compared object's tracking ID. The "*current frame number*" is updated to the "*end frame number*", while the "*start frame number*" remains unchanged which defines when ID tracking started.
2. If any recorded tracking ID is not assigned in the current frame, the *maximum skipped frame number* is incremented by 1. If this value exceeds a predetermined threshold (e.g., 25), no new object will be assigned to the tracking ID, regardless of the distance between points. This decision is based on the assumption that the object is either no longer alive or out of view.
3. If the new objects do not align with any existing tracking ID, create their own distinct tracking IDs.

- **Step 3:** Next, tracking IDs with fewer detections than the threshold defined by "*remove lines less than*" (e.g., 15) are removed, assuming parasites did not survive such brief periods. This eliminates erroneous or incomplete trackings, ensuring the analysis focuses on reliable data.
- **Step 4:** During experiments, Figure 4 was produced by plotting each ROIs midpoints onto a single image, where dense areas indicate high parasite prevalence. Trackings outside these dense boundaries were excluded for robustness. This optional step, interchangeable with Step 3, ensures that even if a tracking ID exceeds the threshold, will be removed if it is not located within dense area(s).

Figure 5 shows tracking results across sequential frames for a video at several-second intervals. Each number represents a unique tracking ID marking the location of a parasite. By following the frame order (from A to D), changes in the ID values can be observed. If the same parasite is tracked in the subsequent frame, the same ID value will appear in that frame as well. For instance, the parasite marked with ID 103 appears in frames A & B but is no longer found or tracked in frames C & D. [2]

Tracking parameters were tuned qualitatively by selecting values that yielded the best performance across various videos. *T. cruzi parasites* defined as FMOs, "*maximum shortest distance*" should not be set too low due to large object displacements between consecutive frames. Too high a value may cause inaccuracies by linking points that belong to different objects. Similarly, parasites must appear in at least a minimum number of frames (e.g., 15) for effective tracking, controlled by the "*remove lines less than*" parameter. Setting this value too high risks losing valid tracks, while too low a value has negligible effect.

4 Experimental Results

Table 1 shows the tracking results for the *T. cruzi* parasites across the 24 videos in our dataset. **Label** is the identifier assigned to each video. The **number of tracked parasites** (NTP) refers to the distinct parasites identified in each video that the algorithm monitored. The **mean of frames** (MF) represents the average

[2]For a better understanding we share the code and some of the processed videos in Github repository.

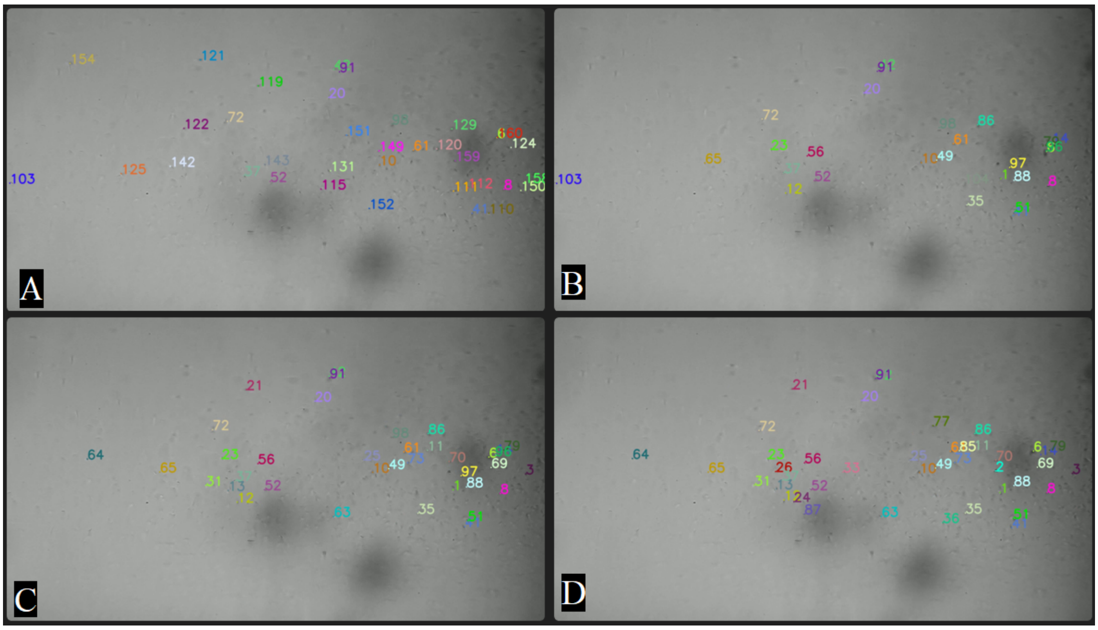


Figure 5: Sample Output of Custom Tracking with FastSam: locations marked with tracking IDs on the sequential images (in order of A-B-C-D) with a few second time intervals, indicate the presence of parasites.

number of frames in which algorithm tracked all detected parasites in the video. In 15 of the 24 videos analyzed, our method utilizing **contour detection/tracing (CT)** segmentation outperformed the other methods by successfully tracking the highest number of parasites. The runner-up was our approach employing FastSAM segmentation, significantly lagging behind the DeepSort and StrongSort methods. Conversely, both techniques, especially StrongSort, significantly outscored our custom method regarding the mean of frames variable. These findings indicate that our method can track a larger quantity of parasites, albeit over a limited number of frames, whereas deep learning techniques are more selective, tracking fewer parasites but for extended time frames.

	Our Method				DeepSort				StrongSort			
	CT		FastSAM		CT		FastSAM		CT		FastSAM	
Label	NTP	MF	NTP	MF	NTP	MF	NTP	MF	NTP	MF	NTP	MF
20	**22**	42.04	21	47.52	3	145.33	4	119.75	3	155.67	2	*179.5*
33	135	80.86	**170**	70.91	40	89.82	49	66.65	23	182.26	21	*208.42*
36	235	80.23	**270**	80.11	74	94.67	70	93.01	62	133.98	58	*137.67*
38	85	60.55	**88**	68.00	61	41.54	55	44.58	21	175.52	15	*201.73*
42	134	66.28	**163**	59.91	39	80.66	42	80.59	26	178.42	24	*178.50*
44	60	57.20	**72**	67.82	50	43.20	45	45.68	18	*164.28*	19	149.73
45	**92**	65.08	89	69.32	55	33.21	44	39.63	14	176.54	12	*196.58*
46	**63**	55.59	53	72.49	13	113.23	12	92.91	10	*215.20*	10	212.80
47	**102**	63.53	97	65.93	43	38.23	40	41.50	15	152.80	8	*175.75*
48	65	50.17	**73**	42.86	12	74.83	13	67.61	10	175.00	8	*180.12*
49	96	55.35	**103**	50.25	18	88.17	19	77.84	11	*188.91*	13	117.46
50	**329**	78.51	208	60.65	92	35.67	44	62.20	26	247.26	20	*253.40*
57	447	103.82	**526**	83.04	197	58.98	215	62.95	48	180.27	29	*230.55*
58	**304**	89.59	234	78.73	59	72.81	43	88.11	36	*171.67*	25	140.56
59	**307**	64.41	253	46.51	40	44.82	31	48.42	13	*135.15*	11	89.36
60	**151**	63.91	138	53.71	35	43.88	27	46.67	12	203.75	9	*213.22*
61	**456**	49.74	175	28.26	37	69.78	10	85.70	11	96.36	5	*137.60*
62	**966**	62.73	470	31.12	113	64.75	52	54.48	95	*190.80*	88	158.04
67	**447**	22.32	335	20.18	31	76.54	32	77.37	26	*153.73*	31	146.84
68	**375**	73.00	343	49.98	40	77.00	44	55.86	43	91.60	30	*95.60*
69	**389**	64.59	311	48.99	53	61.22	43	52.14	40	*98.12*	37	77.38
70	**253**	46.90	215	30.56	29	49.07	22	49.73	11	*228.00*	7	225.28
71	**195**	25.60	140	25.98	12	88.66	12	85.16	11	*191.72*	8	174.00
72	140	64.96	**172**	68.78	96	53.54	101	49.06	44	*179.47*	42	165.78

Table 1: Results for all tested methods. MF values can be converted to seconds using 20 FPS. Bold numbers indicate the highest values for the NTP and italic numbers indicate the highest values for the MF variable.

5 Conclusion

We propose an approach that enables monitoring motile *T. cruzi* parasites in microscopy videos. Current pipeline focuses on movement, contours, and background contrast as primary cues. We encountered challenges in accurately counting the parasites using popular tracking libraries due to their fast movement, small size, and unpredictable behavior. While our approach showed promise, our current model was trained and validated on our specific dataset. Performance could be sensitive to imaging artefacts (e.g., variations in illumination, camera noise) and may not generalize directly to other parasite strains, or microscopy setups. Potential integration with molecular and immunological tools, could create a more robust, multi-modal diagnostic pipeline. The motion-based assay could serve as a rapid, low-cost, and stain-free initial screening tool. Samples flagged as positive could be confirmed with a specific molecular test (e.g., qPCR).

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